Exercise series 8:
3NF

Problem 1:

A) Consider the relation schema Bookings(Title, Theater, City) with functional dependencies:

- Theater → Title
- {Title, City} → Theater

Is the original schema in *3NF* or *BCNF*?

B) If *possible*, give an example of:

1. A relation schema with dependencies that is in *3NF* but not in *BCNF*
2. A relation schema with dependencies that is in *BCNF* but not in *3NF*
3. A relation schema with dependencies that is neither in *BNCF* nor in *3NF*

Problem 1:

A) Consider the relation schema Bookings(Title, Theater, City) with functional dependencies:

- Theater → Title in 3NF but NOT in BCNF
- {Title, City} → Theater in 3NF and in BCNF

Is the original schema in *3NF* or *~~BCNF~~*?

B) If *possible*, give an example of:

1. *3NF* but not in *BCNF* **Done**
2. *BCNF* but not in *3NF* **Impossible** because *BCNF* implies *3NF*
3. *Not BCNF* nor in *3NF* **A → B** and **C → B** **R(A,B,C)**

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,B,C,D → E • E,F → G,H • A,C,D,F → E, G

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,B,C,D → E • E,F → G • E,F → H • A,C,D,F → E • A,C,D,F → G

SINGLE ATTRIBUTE

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,B,C,D → E • E,F → G • E,F → H • A,C,D,F → E • A,C,D,F → G

REMOVE ATTRIBUTES AND CHECK

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,B,C,D → E • E,F → G • E,F → H • A,C,D,F → E • A,C,D,F → G

REMOVE ATTRIBUTES AND CHECK

$\{A,B,C\}^{+}= \{A,B,C\}$

$\{A,B,D\}^{+}= \{A,B,D\}$

$\{A,C,D\}^{+}= \{A,B,C,D,E\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,C,D → E • E,F → G • E,F → H • A,C,D,F → E • A,C,D,F → G

REMOVE ATTRIBUTES AND CHECK

$\{A,C\}^+ = \{A,B,C\}$

$\{A,D\}^+ = \{A,B,D\}$

$\{C,D\}^+ = \{C,D\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A $\rightarrow$ B • A,C,D $\rightarrow$ E • E,F $\rightarrow$ G • E,F $\rightarrow$ H • A,C,D,F $\rightarrow$ E • A,C,D,F $\rightarrow$ G

REMOVE ATTRIBUTES AND CHECK

$\{E\}^+ = \{E\}$

$\{F\}^+ = \{F\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,C,D → E • E,F → G • E,F → H • A,C,D,F → E • A,C,D,F → G

REMOVE ATTRIBUTES AND CHECK

$\{E\}^{+} = \{E\}$

$\{F\}^{+} = \{F\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,C,D → E • E,F → G • E,F → H • A,C,D,F → E • A,C,D,F → G

REMOVE ATTRIBUTES AND CHECK

$\{A,C,D\}^{+} = \{A,B,C,D,E\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,C,D → E • E,F → G • E,F → H • A,C,D → E • A,C,D,F → G

REMOVE ATTRIBUTES AND CHECK

$\{A,C\}^+ = \{A,B,C\}$

$\{A,D\}^+ = \{A,B,D\}$

$\{C,D\}^+ = \{C,D\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• $A \rightarrow B$ • $A,C,D \rightarrow E$ • $E,F \rightarrow G$ • $E,F \rightarrow H$ • $A,C,D \rightarrow E$ • $A,C,D,F \rightarrow G$

REMOVE ATTRIBUTES AND CHECK

$\{A,C,D\}^+ = \{A,B,C,D,E\}$

$\{A,C,F\}^+ = \{A,B,C,F\}$

$\{A,D,F\}^+ = \{A,B,D,F\}$

$\{C,D,F\}^+ = \{B,C,D\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,C,D → E • E,F → G • E,F → H • A,C,D → E • A,C,D,F → G

REMOVE FDs

$\{A\}^{+} = \{A\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,C,D → E • E,F → G • E,F → H • A,C,D → E • A,C,D,F → G

REMOVE FDs

$\{A,C,D\}^+ = \{A,B,C,D,E\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A $\rightarrow$ B • E,F $\rightarrow$ G • E,F $\rightarrow$ H • A,C,D $\rightarrow$ E • A,C,D,F $\rightarrow$ G

REMOVE FDs

$\{E,F\}^{+} = \{E,F,H\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A $\rightarrow$ B • E,F $\rightarrow$ G • E,F $\rightarrow$ H • A,C,D $\rightarrow$ E • A,C,D,F $\rightarrow$ G

REMOVE FDs

$\{E,F\}^{+} = \{E,F,G\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • E,F → G • E,F → H • A,C,D → E • A,C,D,F → G

REMOVE FDs

$\{A,C,D\}^{+} = \{A,B,C,D\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • E,F → G • E,F → H • A,C,D → E • A,C,D,F → G

REMOVE FDs

$\{A,C,D,F\}^+ = \{A,B,C,D,E,F,G,H\}$

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • E,F → G • E,F → H • A,C,D → E

RECOMPOSE

R(A,B,C,D,E,F,G,H)
• A → B • E,F → G,H • A,C,D → E

Problem 3: Decompose to 3NF

a) Which of the following tables are already in 3NF?

b) Decompose the other tables in 3NF using the algorithm

R(A,B,C,D)	R(A,B,C,D)	R(A,B,C,D)	R(A,B,C,D)	R(A,B,C,D,E)	R(A,B,C,D,E)
• A,B → C	• B → C,D	• A,B → C	• A → B	• A,B → C	• A,B → C
• C → D		• B,C → D	• B → C	• D,E → C	• C → D
• D → A		• C,D → A	• C → D	• B → D	• D → B,E
		• A,D → B	• D → A		

Problem 3: Decompose to 3NF

a) Which of the following tables are already in 3NF?

R(A,B,C,D)

- A,B → C
- C → D
- D → A

Keys:

- {A,B}
- {B,C}
- {B,D}

R(A,B,C,D)

- B → C,D

Keys:

- {A,B}

R(A,B,C,D)

- A,B → C
- B,C → D
- C,D → A
- A,D → B

Keys:

- {A,B}
- {B,C}
- {C,D}
- {A,D}

R(A,B,C,D)

- A → B
- B → C
- C → D
- D → A

Keys:

- {A}
- {B}
- {C}
- {D}

R(A,B,C,D,E)

- A,B → C
- D,E → C
- B → D

Keys:

- {A,B,E}

R(A,B,C,D,E)

- A,B → C
- C → D
- D → B,E

Keys:

- {A,B}
- {A,C}
- {A,D}

Problem 3: Decompose to 3NF

b) Decompose the other tables in 3NF using the algorithm

R(A,B,C,D)
• B → C,D

R(A,B,C,D,E)
• A,B → C • D,E → C • B → D

R(A,B,C,D,E)
• A,B → C • C → D • D → B,E

Problem 3: Decompose to 3NF

b) Decompose the other tables in 3NF using the algorithm

R(A,B,C,D)
• $B \rightarrow C$ • $B \rightarrow D$

R(A,B,C,D,E)
• $A,B \rightarrow C$ • $D,E \rightarrow C$ • $B \rightarrow D$

R(A,B,C,D,E)
• $A,B \rightarrow C$ • $C \rightarrow D$ • $D \rightarrow B$ • $D \rightarrow E$

- Single attribute on the right

Problem 3: Decompose to 3NF

b) Decompose the other tables in 3NF using the algorithm

R(A,B,C,D)
• B → C • B → D

R(A,B,C,D,E)
• A,B → C • D,E → C • B → D

R(A,B,C,D,E)
• A,B → C • C → D • D → B • D → E

- Already basis

Problem 3: Decompose to 3NF

b) Decompose the other tables in 3NF using the algorithm

R(A,B,C,D)
• B → C, D

R(A,B,C,D,E)
• A,B → C • D,E → C • B → D

R(A,B,C,D,E)
• A,B → C • C → D • D → B, E

- Recompose

Problem 3: Decompose to 3NF

b) Decompose the other tables in 3NF using the algorithm

R(A,B,C,D)
• B → C, D

R(A,B,C,D,E)
• A,B → C • D,E → C • B → D

R(A,B,C,D,E)
• A,B → C • C → D • D → B, E

- Relations from FDs

R(B,C,D)

R(A,B,C)
R(C,D,E)
R(B,D)

R(A,B,C)
R(C,D)
R(D,B,E)

Problem 3: Decompose to 3NF

b) Decompose the other tables in 3NF using the algorithm

R(A,B,C,D)
• B → C, D

R(A,B,C,D,E)
• A,B → C
• D,E → C
• B → D

R(A,B,C,D,E)
• A,B → C
• C → D
• D → B, E

- Add key if needed

R(B,C,D)
R(A,B)

R(A,B,C)
R(C,D,E)
R(B,D)
R(A,B,E)

R(A,B,C)
R(C,D)
R(D,B,E)

Problem 4: Lossless and dep. preserving

Let R(A,B,C,D,E) be decomposed into relations R_1(A,B,C), R_2(B,C,D), and R_3(A,C,E).

For each of the following sets of FDs check if the decomposition preserves dependencies and/or is lossless.

- $B \rightarrow E$ and $CE \rightarrow A$.
- $AC \rightarrow E$ and $BC \rightarrow D$.
- $A \rightarrow D$, $D \rightarrow E$, and $B \rightarrow D$.
- $A \rightarrow D$, $CD \rightarrow E$, and $E \rightarrow D$.

Problem 4: Lossless and dep. preserving

Let R(A,B,C,D,E) be decomposed into relations R_1(A,B,C), R_2(B,C,D), and R_3(A,C,E).

For each of the following sets of FDs check if the decomposition preserves dependencies and/or is lossless.

- $B \rightarrow E$ and $CE \rightarrow A$.
- $AC \rightarrow E$ and $BC \rightarrow D$.
- $A \rightarrow D$, $D \rightarrow E$, and $B \rightarrow D$.
- $A \rightarrow D$, $CD \rightarrow E$, and $E \rightarrow D$.

Problem 4: Lossless and dep. preserving

Let R(A,B,C,D,E) be decomposed into relations R_1(A,B,C), R_2(B,C,D), and R_3(A,C,E).

For each of the following sets of FDs check if the decomposition preserves dependencies and/or is lossless.

- $B \rightarrow E$ and $CE \rightarrow A$.
- $AC \rightarrow E$ and $BC \rightarrow D$.
- $A \rightarrow D$, $D \rightarrow E$, and $B \rightarrow D$.
- $A \rightarrow D$, $CD \rightarrow E$, and $E \rightarrow D$.

	A	B	C	D	E
R_1	a	b	c	d_1	e_1
R_2	a	b	c	d	e_1
R_3	a	b_3	c	d_3	e

Problem 4: Lossless and dep. preserving

Let $R(A,B,C,D,E)$ be decomposed into relations $R_1(A,B,C)$, $R_2(B,C,D)$, and $R_3(A,C,E)$.

For each of the following sets of FDs check if the decomposition preserves dependencies and/or is lossless.

- $B \rightarrow E$ and $CE \rightarrow A$.
- $AC \rightarrow E$ and $BC \rightarrow D$.
- $A \rightarrow D$, $D \rightarrow E$, and $B \rightarrow D$.
- $A \rightarrow D$, $CD \rightarrow E$, and $E \rightarrow D$.

	A	B	C	D	E
R_1	a	b	c	d	e
R_2	a_2	b	c	d	e_2
R_3	a	b_3	c	d_3	e

Problem 4: Lossless and dep. preserving

Let R(A,B,C,D,E) be decomposed into relations R_1(A,B,C), R_2(B,C,D), and R_3(A,C,E).

For each of the following sets of FDs check if the decomposition preserves dependencies and/or is lossless.

- $B \rightarrow E$ and $CE \rightarrow A$.
- $AC \rightarrow E$ and $BC \rightarrow D$.
- $A \rightarrow D$, $D \rightarrow E$, and $B \rightarrow D$.
- $A \rightarrow D$, $CD \rightarrow E$, and $E \rightarrow D$.

	A	B	C	D	E
R_1	a	b	c	d	e
R_2	a_2	b	c	d	e_2
R_3	a	b_3	c	d_1	e

Problem 4: Lossless and dep. preserving

Let R(A,B,C,D,E) be decomposed into relations R_1(A,B,C), R_2(B,C,D), and R_3(A,C,E).

For each of the following sets of FDs check if the decomposition preserves dependencies and/or is lossless.

- $B \to E$ and $CE \to A$.
- $AC \to E$ and $BC \to D$.
- $A \to D$, $D \to E$, and $B \to D$.
- $A \to D$, $CD \to E$, and $E \to D$.

	A	B	C	D	E
R_1	a	b	c	d_1	e
R_2	a_2	b	c	d	e_2
R_3	a	b_3	c	d_1	e

Problem 5: Dependency preserving

Check if the following decomposition are dep. preserving.

R(A,B,C,D)
• $A \rightarrow B$ • $B \rightarrow C$ • $C \rightarrow D$ • $D \rightarrow A$

decomposed into $R_1(A,C)$ $R_2(C,D)$ $R_3(B,D)$

R(A,B,C,D,E,F,G)
• $A \rightarrow E$ • $B,E \rightarrow F$ • $C,F \rightarrow G$ • $A,G \rightarrow D$ • $A,B,C \rightarrow D$

decomposed into $R_1(A,D,E,G)$ $R_2(B,E,F)$ $R_3(C,F,G)$

Problem 5: Dependency preserving

R(A,B,C,D)
• A → B • B → C • C → D • D → A

Without the algorithm

R_1(A,C)
• A → C • C → A

R_2(C,D)
• C → D • D → C

R_3(B,D)
• B → D • D → B

- A → C → D → B
- B → D → C
- C → D
- D → C → A

R(A,B,C,D,E,F,G)
• A → E • B,E → F • C,F → G • A,G → D • A,B,C → D

With the algorithm

X :={A,B,C}

R_1(A,D,E,G)	R_2(B,E,F)	R_3(C,F,G)

Problem 5: Dependency preserving

R(A,B,C,D)

- A → B
- B → C
- C → D
- D → A

Without the algorithm

R_1(A,C)

- A → C
- C → A

R_2(C,D)

- C → D
- D → C

R_3(B,D)

- B → D
- D → B

- A → C → D → B
- B → D → C
- C → D
- D → C → A

R(A,B,C,D,E,F,G)

- A → E
- B,E → F
- C,F → G
- A,G → D
- A,B,C → D

With the algorithm

X :={A,B,C} U {E}

R_1(A,D,E,G) | **R_2(B,E,F)** | **R_3(C,F,G)**

Problem 5: Dependency preserving

R(A,B,C,D)

- A → B
- B → C
- C → D
- D → A

Without the algorithm

R_1(A,C)

- A → C
- C → A

R_2(C,D)

- C → D
- D → C

R_3(B,D)

- B → D
- D → B

- A → C → D → B
- B → D → C
- C → D
- D → C → A

R(A,B,C,D,E,F,G)

- A → E
- B,E → F
- C,F → G
- A,G → D
- A,B,C → D

With the algorithm

X :={A,B,C,E} U {F}

R_1(A,D,E,G) **R_2(B,E,F)** **R_3(C,F,G)**

Problem 5: Dependency preserving

R(A,B,C,D)

- $A \rightarrow B$
- $B \rightarrow C$
- $C \rightarrow D$
- $D \rightarrow A$

Without the algorithm

R_1(A,C)

- $A \rightarrow C$
- $C \rightarrow A$

R_2(C,D)

- $C \rightarrow D$
- $D \rightarrow C$

R_3(B,D)

- $B \rightarrow D$
- $D \rightarrow B$

- $A \rightarrow C \rightarrow D \rightarrow B$
- $B \rightarrow D \rightarrow C$
- $C \rightarrow D$
- $D \rightarrow C \rightarrow A$

R(A,B,C,D,E,F,G)

- $A \rightarrow E$
- $B,E \rightarrow F$
- $C,F \rightarrow G$
- $A,G \rightarrow D$
- $A,B,C \rightarrow D$

With the algorithm

$X := \{A,B,C,E,F\} \cup \{G\}$

R_1(A,D,E,G) **R_2(B,E,F)** **R_3(C,F,G)**

Problem 5: Dependency preserving

R(A,B,C,D)

- A → B
- B → C
- C → D
- D → A

Without the algorithm

R_1(A,C)

- A → C
- C → A

R_2(C,D)

- C → D
- D → C

R_3(B,D)

- B → D
- D → B

- A → C → D → B
- B → D → C
- C → D
- D → C → A

R(A,B,C,D,E,F,G)

- A → E
- B,E → F
- C,F → G
- A,G → D
- A,B,C → D

With the algorithm

X :={A,B,C,E,F,G} U {D}

R_1(A,D,E,G) **R_2(B,E,F)** **R_3(C,F,G)**

Problem 5: Dependency preserving

R(A,B,C,D)

- A → B
- B → C
- C → D
- D → A

Without the algorithm

R_1(A,C)

- A → C
- C → A

R_2(C,D)

- C → D
- D → C

R_3(B,D)

- B → D
- D → B

- A → C → D → B
- B → D → C
- C → D
- D → C → A

R(A,B,C,D,E,F,G)

- A → E
- B,E → F
- C,F → G
- A,G → D
- A,B,C → D

With the algorithm

X :={A,B,C,E,F,G} U {D}

R_1(A,D,E,G) **R_2(B,E,F)** **R_3(C,F,G)**